

HAT-P-32b

R-0132

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_230112 · created 2026-07-16T06:22:27+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459954.5704 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.053
Transit depth (Rp/Rs)^2	3.41 +/- 1.97 [%]
Orbital Inclination (inc)	87.55 +/- 1.71
Airmass coefficient 1 (a1)	0.944 +/- 0.043
Airmass coefficient 2 (a2)	0.06 +/- 0.33
Scatter in the residuals of the lightcurve fit is	4.64 %
Best Comparison Star	#3 - [89, 288]
Optimal Method	PSF photometry
Transit Duration (day)	0.0617 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.185 ± 0.053 vs 0.1489 (deviation 0.68σ)
Duration fit vs published	0.0617 d vs 0.1304 d (deviation 13.47σ)
Mid-time O-C	-3092.6 min
Depth SNR	3.5
Residual scatter	4.64 %

Light curve

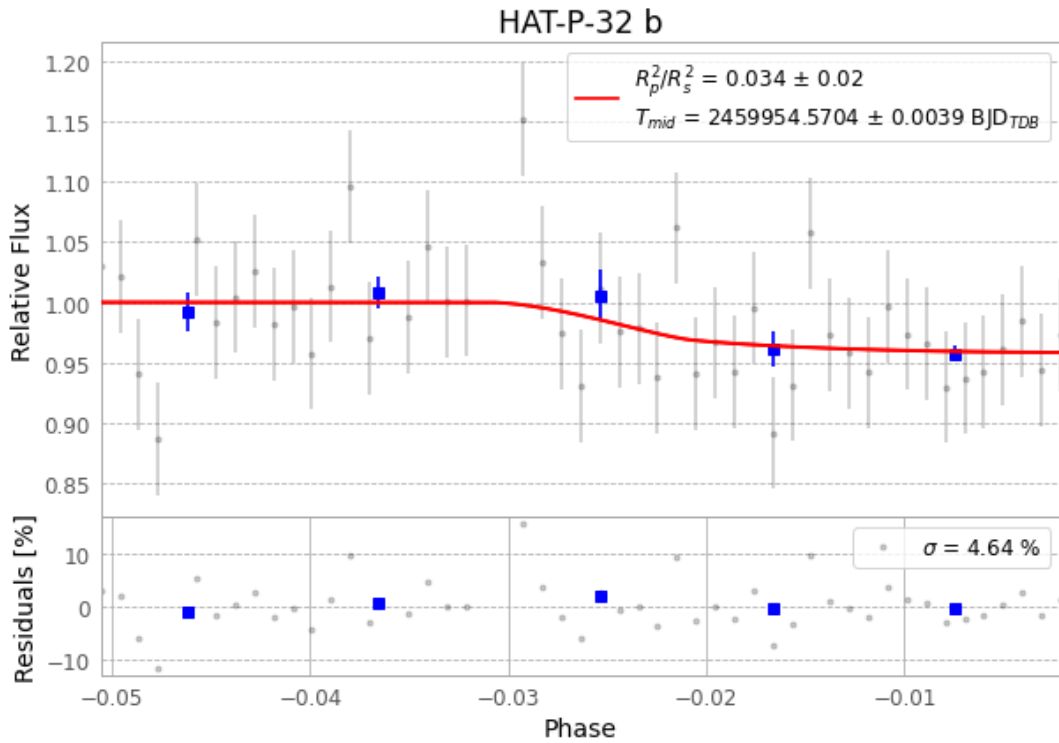


Figure 1 — FinalLightCurve_HAT-P-32 b_2023-01-11.png (SHA-256 c9f6bf0e43bc21d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [480, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e0404b0910234007...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

51 FITS frames (33 MB), HATP-32230112023615.FITS → HATP-32230112050616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-230112.log (e8ca6efbcbcc...), FinalLightCurve_HAT-P-32 b_2023-01-11.png (c9f6bf0e43bc...), FinalLightCurve_HAT-P-32 b_2023-01-11.csv (b7fb05b14ca6...)

Compute resources

Wall time 205.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 06:22Z.